

# API Systems - Task 3 Evaluation

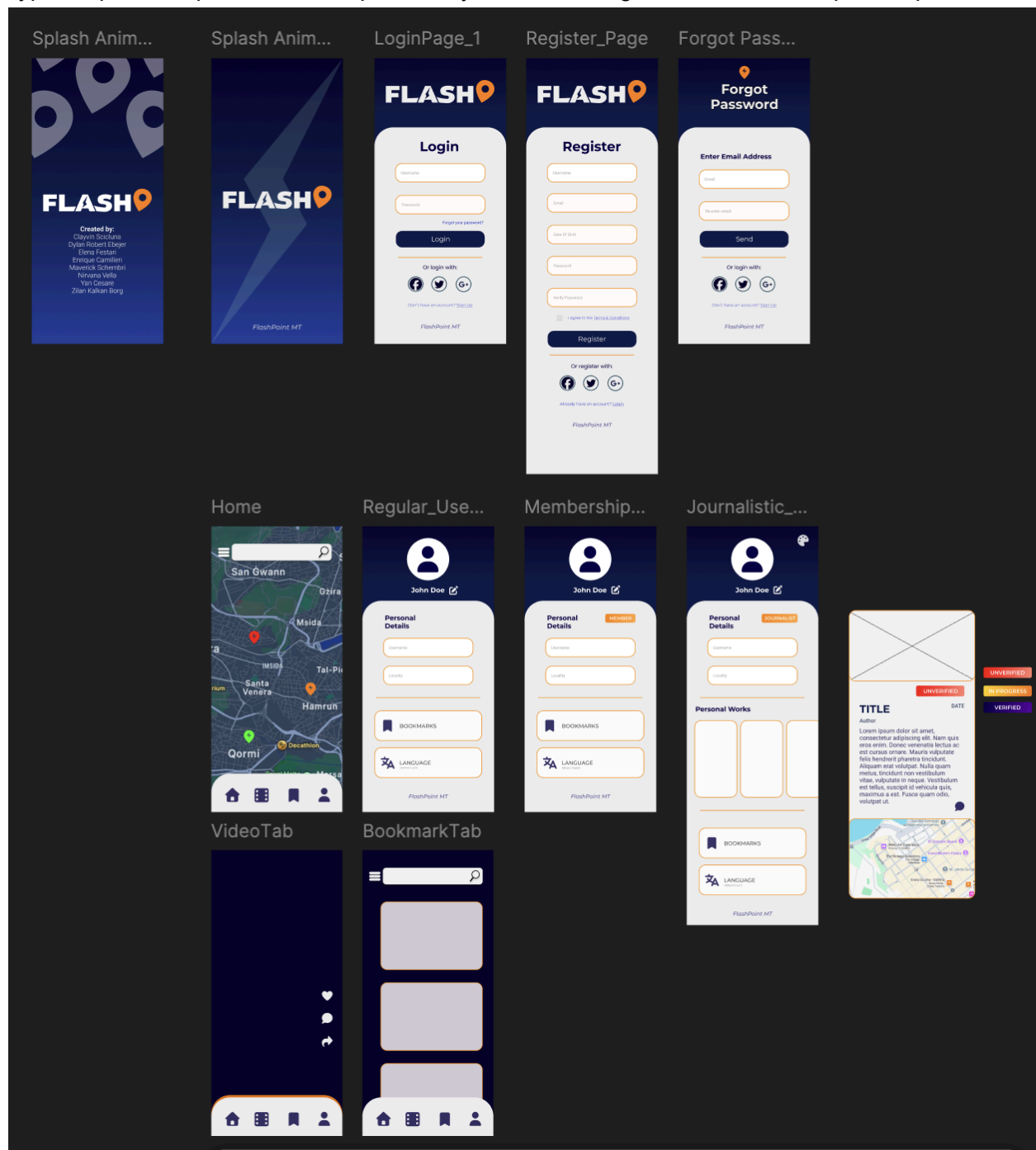
In the beginning of the semester, the project initially was going to be a game-based API with no real direction. After some brainstorming there was a direction forming around creating achievements and a leaderboard system, but frankly the burnout of such a challenging semester brought the conclusion that maybe a new project would be better, and experimenting with something entirely new could be more interesting as well as more beneficial in the long run. After some discussion with the entrepreneurship group we had settled on creating an application where the main functionality would be an interactive map. The concept was that the user can navigate specific areas within the Maltese islands and see the news relevant to their chosen area. The users would be able to filter out specific types of news, which was not fully implemented as of yet, but the categories are implemented within the database for future development ease. Essentially it would be a journalistic application where the user can either view the news of their choosing or create news articles under a specific membership called the "Journalistic Membership". With such an application, implementing APIs is much more interesting and diverse compared to the original game idea, as the range of endpoints required covers authentication, content management, user interaction and verification workflows all at once.

## Design

When it comes to the design of this application, there was initially quite a bit of brainstorming and multiple iterations, considering the goal was to create a user friendly application aimed towards Gen Z. The design had to be modern, clean and in keeping with the theme of journalism, something that felt credible and trustworthy while still being visually engaging enough to compete with the social media platforms that the target audience uses daily. Overall, this aspect was the most interesting and entertaining part of the project for me personally, as I wanted to test and develop the graphic design skills I have been building throughout my internship experience.

The designs were first done on Figma to establish a clear direction before any development began. This allowed the team to align on the visual language early, settling on a dark navy and orange colour palette that references the urgency of breaking news while still feeling modern and fresh. Multiple screens were designed including the splash animation, login page, registration page, forgot password flow, the home map page, and three variations of the user profile depending on membership type. Having these Figma designs as a reference made the development process significantly more structured, as there was always a clear visual target to work towards rather than making design decisions mid-development which tends to slow things down. Once the designs were finalised they were implemented within the Ionic Angular application, which required translating the Figma components into actual HTML, SCSS and

TypeScript was a process that improved my understanding of frontend development quite a lot.



## Development

The application was developed using Angular CLI and Ionic for the frontend and user interface, with PHP handling the backend functionality and API endpoints. MAMP was used as the local

server environment on macOS, Postman was used for API testing, and MySQL managed through phpMyAdmin handled the database. Initially when starting the development it was mainly the frontend which required the most attention, using TypeScript extensively which was a bit hard to get used to at first. However, since TypeScript shares some similarities with C# in terms of its type system and object-oriented approach, it became more manageable as time went on and confidence grew with the language.

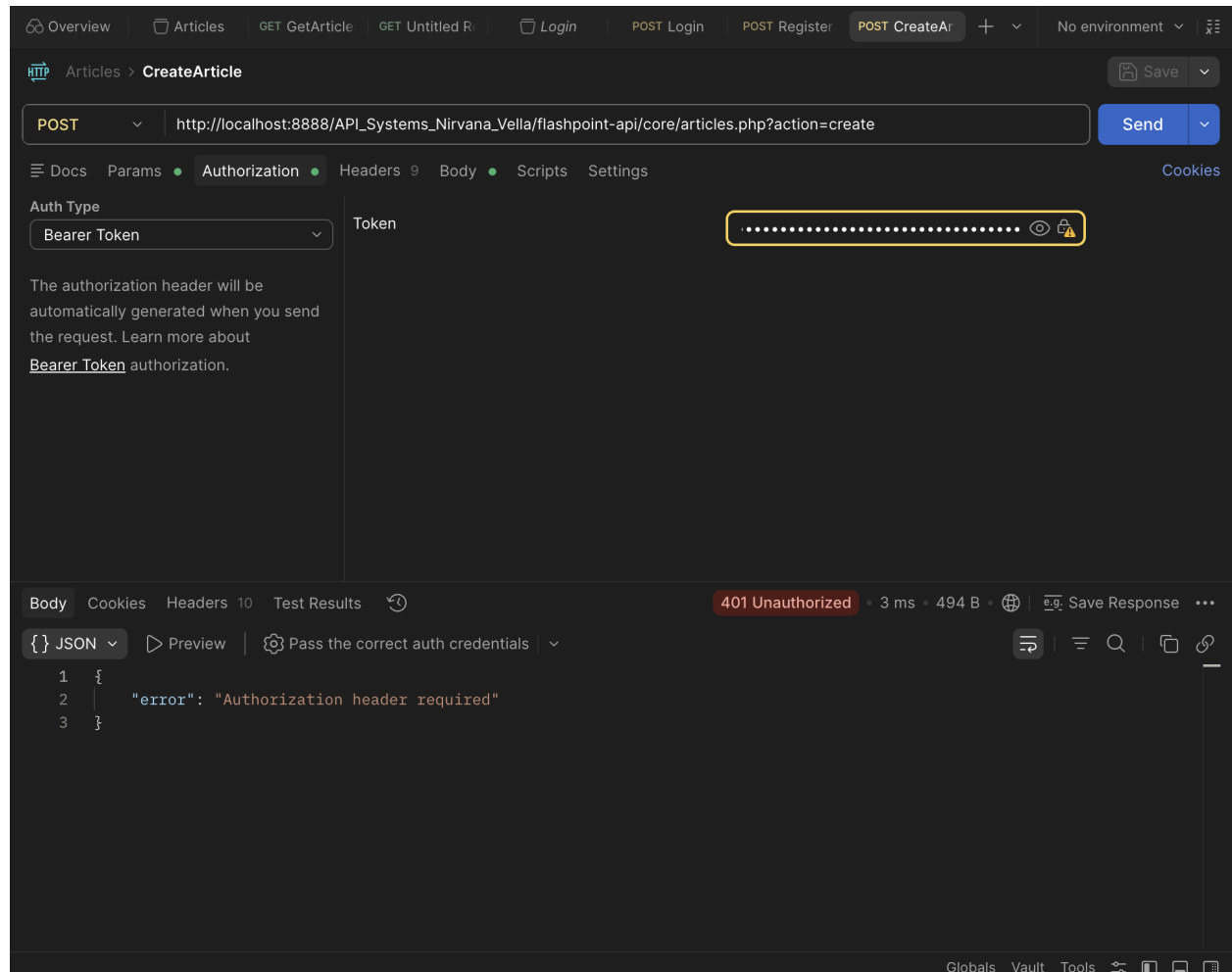
There were some technical challenges throughout the development process. One of the more notable ones was connecting the login system with the interactive map, the routing between pages was not working correctly at first, returning an Angular routing error that prevented navigation after a successful login. This was eventually resolved by correcting a duplicate path definition in the tab routing configuration. Another challenge was the MAMP environment itself, which required specific port configurations for both Apache and MySQL that differed from the XAMPP setup the team had initially used. The database connection string needed the MySQL port explicitly set to 8889 and the host updated to 127.0.0.1, which was not immediately obvious and took some troubleshooting to identify.

The overall implementation of the API endpoints was relatively straightforward once the environment was configured correctly. For my side of the project I handled the articles, comments and bookmarks endpoints. The articles endpoint covers listing articles with filtering by status, category and verification state, creating new articles restricted to journalist and admin roles, and updating the verification status through a structured flow of unverified to in progress to verified. The comments endpoint allows any authenticated user to post comments under articles while the GET endpoint remains public. The bookmarks endpoint handles adding, listing and removing bookmarks for the logged in user. OAuth 2.0 was implemented for the login and token-based authentication, which was admittedly a bit of a headache as it was not something covered directly in class. After some trial and error we managed to implement a JWT-based Bearer token system that is generated on login and validated on every protected endpoint request. Yan handled the login, user management, membership and profile side of the API.

## Testing

The testing was primarily conducted through Postman, where collections were created for each resource group: articles, bookmarks and comments. Most of the endpoints worked immediately and smoothly once the database connection and routing were resolved. However, when it came to the create article endpoint there were some difficulties. The endpoint requires the authenticated user to have either a journalist or admin role in order to submit an article, but during testing the logged in user was not being recognised as a journalist despite being registered with that role. After some investigation it turned out the issue was related to the token validation in the config file not correctly reading the authorisation header in the MAMP environment. Once the header fallback checks were added to the requireAuth function the role was correctly extracted from the token and the endpoint began working as expected. After that, all endpoints were tested with both success cases and error cases: including 401 unauthorised,

403 forbidden, 404 not found and 409 conflict responses, and everything worked according to plan.



## Future Improvements

There are several areas where this application could be significantly improved given more time and resources. Firstly, security measures could be considerably strengthened. The current implementation still has relatively basic security protections which would not withstand every potential threat, particularly in an online environment where many different users with varying intentions could be interacting with the system. Token invalidation on logout and password change would also improve the security posture considerably, as currently issued tokens remain valid until they expire regardless of whether the user has logged out.

Beyond security, the filtering functionality on the map could be fully implemented to allow users to filter news pins by category, date range and verification status in real time. The verification workflow could be expanded to include notification emails to journalists when their articles move through the review stages. The social login functionality via Google, Facebook and Twitter was

partially planned but not fully completed during this semester, and completing that integration would significantly lower the barrier to entry for new users. Finally, a media upload feature allowing journalists to attach images and video clips to their articles would greatly enhance the richness of the content on the platform and bring it closer to the original vision of a credible, modern journalistic application for Maltese citizens.